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CAPSTONE PROJECT

E-commerce Sales Analysis using Power BI

PRESENTED BY

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OUTLINE:

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PROBLEM STATEMENT:

In the rapidly growing e-commerce industry, a large amount of sales data is generated daily. Analyzing this data manually is time-consuming and inefficient. Businesses need an effective way to track sales performance, identify trends, and make data-driven decisions. There is a need for an interactive system that can visualize sales, profit, and customer insights in an easy-to-understand manner.

PROPOSED SOLUTION:

The proposed system is an interactive dashboard developed using Power BI. It analyzes e-commerce sales data and provides meaningful insights such as total sales, profit, customer behavior, and regional performance. The dashboard allows users to filter data dynamically and supports better decision-making.

SYSTEM APPROACH:

- **Tool Used: Power BI Desktop**
- **Data Source: CSV file (Superstore dataset)**
- **Data Cleaning: Power Query Editor**
- **Data Modeling: Relationships between tables**
- **Visualization: Charts, graphs, maps**
- **DAX: Used for calculations like Sales, Profit, and Profit Margin**

REPORT & ANALYSIS:

Data Report:

The dataset consists of e-commerce sales records including order date, sales amount, profit, quantity, product category, customer details, and region. Data cleaning and preprocessing were performed using Power Query to handle missing values and ensure consistency.

Analysis Performed:

- **Time-based analysis to identify sales trends**
- **Category and sub-category performance evaluation**
- **Region-wise sales and profit comparison**
- **Identification of top-selling and low-performing products**

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Key Insights:

- **Some product categories contribute significantly to overall sales**
- **Certain regions generate higher profit margins**
- **A small number of products drive a large portion of revenue**
- **Seasonal trends affect sales performance**

Functions (DAX) Used:

- **Total Sales = SUM(Sales)**
- **Total Profit = SUM(Profit)**
- **Profit Margin = DIVIDE(Profit, Sales, 0)**
- **Total Orders = COUNT(Order ID)**
- **Average Sales = AVERAGE(Sales)**

RESULT:

The dashboard provides:

- **Clear visualization of sales and profit trends**
- **Identification of top-performing products**
- **Region-wise sales insights**
- **Easy comparison of different categories**

E-commerce Sales Dashboard

Category
- Furniture
- Office Supplies
- Technology

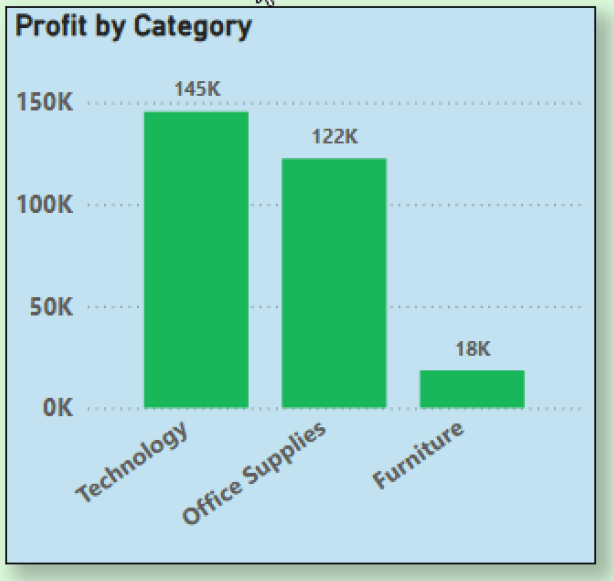
Region
- Central
- East
- South
- West

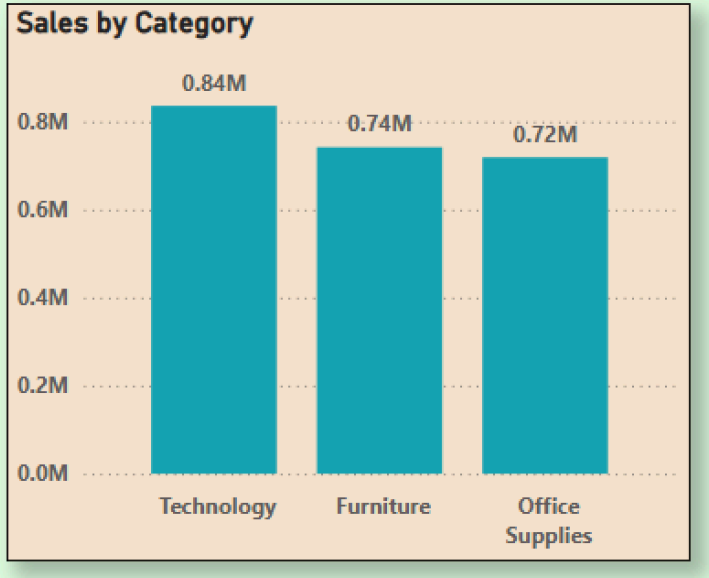
Order Date
- 1/1/2017
- 1/10/2014
- 1/10/2015
- 1/10/2016

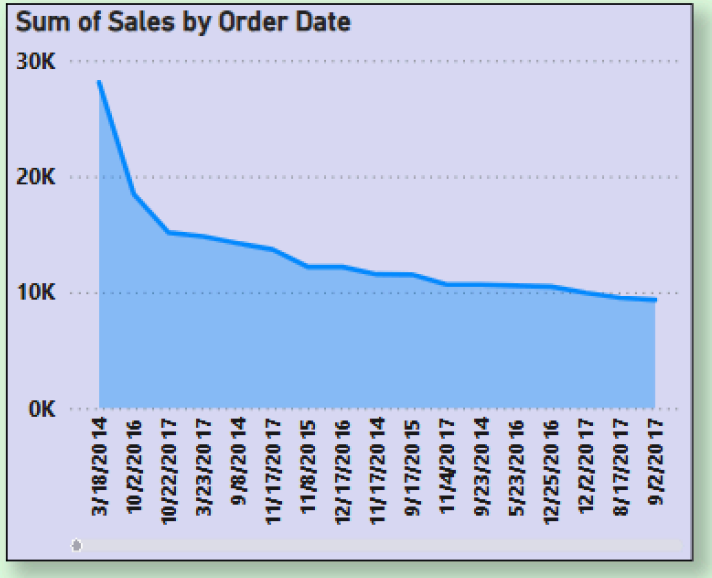
2.30M
Total Sales

286.40K
Total Profit

10K
Total Orders







Screenshot

CONCLUSION:

The Power BI dashboard successfully transforms raw sales data into meaningful insights. It helps businesses understand their performance, identify trends, and make better decisions. The project demonstrates the effectiveness of data visualization in solving real-world business problems.

FUTURE SCOPE:

- Integration with real-time data sources
- Use of AI for sales forecasting
- Mobile-friendly dashboard
- Integration with business systems

REFERENCES:

- Kaggle Dataset (Superstore Sales Data)
- Microsoft Power BI Documentation
- Online tutorials and learning resources

- GitHub Link : https://github.com/raushankumar44/Ecommerce-Sales-Dashboard-PowerBI

Thank You